MILITARY SPECIFICATION

BLOWER, BRUSHLESS, ANTI-EXPOSURE SUIT VENTILATION, GENERAL SPECIFICATION FOR

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers anti-exposure suit ventilation brushless blowers, operable from 28 volt dc aircraft power systems,

2. APPLICABLE DOCUMENTS

The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy; Processes for Corrosion, Protection of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment

FSC 1660

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SPECIFICATIONS

Military (Cont'd)

MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-P-7936	Parts, and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coating, for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-195	Marking of Connections for Electrical Connections
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS17269	Blower, D.C. Brushless, Continuous Duty, Garment Ventilation, Mark 5, Anti-exposure Suit
MS33543	Criteria Temperature and Altitude Range Self Cooled Electric Equipment
MS33586	Metals, Definition of Dissimilar

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality control representative. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Qualification - The blowers furnished under this specification shall be a product which has been tested, and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Materials - Materials used in the manufacture of units shall be of high quality, suitable for the purpose, and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used provided the specifications are approved by the Qualifying Activity and contain provisions for adequate tests. The use of the contractor's specifications will not constitute waiver of Government inspection.

3.2.1 Metals - Metals shall be of a corrosion-resistant type or suitably processed to resist corrosion.

3.2.2 Dissimilar metals - Unless otherwise suitably protected, dissimilar metals, such as brass, copper, or steel, shall not be used in intimate contact with magnesium, aluminum, or their alloys. When protection is used, it shall be of such type that a low-impedance path is offered to radio-frequency currents. Dissimilar metals are defined in MS33586.

3.2.3 Selection of materials - Specifications and standards of all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.2.3.1 Standard parts - Standard parts (MS or AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Any standard part, of the same part number, shall be completely interchangeable as applied within the unit. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the standard parts (MS or AN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list, and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and construction - Blowers shall be of the centrifugal fan design, powered by an integrally mounted brushless motor, capable of operating in accordance with the requirements of this specification and MS17269(AS) when installed in any position.

3.3.1 Construction - Blowers shall be constructed in accordance with the requirements of this specification and MS17269(AS).

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3.3.2 Operating conditions - The unit shall be designed to furnish cooling air continuously under the following conditions or natural combination thereof when the input voltage is varied over the range permitted by MS17269(AS).

3.3.2.1 Temperature/altitude - The unit shall meet the performance requirements at any altitude from sea level to 20,000 ft. in an ambient temperature of 10°C to 43°C at any of these altitudes. The unit shall not be damaged by continuous operation under the temperature/altitude conditions within the range established by Curve II of MS33543 at altitudes from 20,000 to 40,000 feet. Barometric pressures may remain constant or may vary at a rate of 1.5 inch Hg/sec.

3.3.2.2 Ambient temperature - Ambient temperature shall be defined on the applicable drawing or detail specification for any particular unit.

3.3.2.3 Humidity - Relative humidity ranging up to 100 percent, including conditions wherein condensation will take place on the equipment.

3.3.2.4 Sand and dust resistance - Under conditions of airborne sand and dust particles.

3.3.2.5 Salt spray - Atmosphere containing salt-laden moisture.

3.3.2.6 Fungus - When exposed to fungus growth in accordance with Procedure 1 of MIL-E-5272.

3.3.2.7 Overspeed - The unit shall be capable of operating at no load at an input voltage of 35 volts dc when tested as specified in 4.5.10.

3.3.2.8 Shock - When subjected to a shock condition as required by the Shock tests, Procedure V, of MIL-E-5272.

3.3.2.9 Input power - The input power shall be in accordance with MIL-STD-704. The type power used shall be as specified in MS17269(AS).

3.3.2.10 Duty cycle - The duty cycle shall be continuous operation.

3.3.2.11 Temperature rise - The maximum temperature increase of the air through the blower shall be less than 13°F.

3.3.3 Lubrication - Units shall not require lubrication during 1500 hours of service life.

3.3.4 Air Flow Output - The airflow output shall be not less than that specified on MS17269(AS).

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3.3.5 Electrical connections - External termination of the motor wiring shall be in accordance with MIL-STD-195, The motor mounting provisions shall provide a low-impedance path for radio-interference currents, but shall not be used to complete motor electrical circuits in lieu of a lead. Wherever practicable, positive connections which do not depend upon insulation in compression shall be used. Suitable shielding shall be provided for remote filtering if specified in the detail specification.

3.3.5.1 Reverse polarity - The blower shall not be damaged by the application of reversed voltage polarity for 10 minutes.

3.3.6 Armature - Motors shall be so constructed that the armature is readily removed without the use of special tools.

3.3.7 Protection - All ventilating openings in the motor housing shall be of such size as not to permit passage of a 5/16-inch diameter rod. No area on the unit with all covers in place shall be exceptionally vulnerable to damage.

3.3.8 Life - The unit shall be so designed that when operating under any temperature or altitude conditions indicated by MS17279(AS) the service life of the unit without maintenance shall be no less than 1500 hours.

3.3.9 Surge - The blower shall not surge over the operating range from 0 to 16 cfm.

3.3.10 Fragment protection - The impeller housing shall be designed to contain all fragments in the event of impeller failure.

3.3.11 Explosion-proof - The blower motor shall be explosion-proof as defined in 4.5.23 when specifically required by the detail specification or drawing.

3.3.12 Radio interference - Conducted and radiated interference produced by the operation of the unit shall not exceed the requirements of MIL-I-6181.

3.3.13 Ripple voltage - The peak voltage shall be not more than 1.5 volts.

3.3.14 Transient voltage - The blower shall not be damaged by application of transient dc voltage as shown in MIL-STD-704.

3.3.15 Acoustical noise - Acoustical noise measurements taken with a microphone one foot from the blower housing at any point shall not exceed 90 decibels.

3.4 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation and performance. Changes in manufacturer's part numbers shall be goverened by the drawing number requirements of MIL-D-1000.

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3.5 Finish -

3.5.1 Anodizing - When used, anodic treatment of aluminum alloy parts shall be in accordance with MIL-A-8625. The aluminum-oxide film deposited by this treatment shall be removed from the actual contact area of all surfaces required to act as a path for electrical current and from the local area under screws, nuts, or the like used for assembly or mounting purposes, to provide an adequate bonding connection.

3.5.2 Chemical films - Chemical films in accordance with MIL-C-5541, or approved equivalent, may be used when the motor component parts are not subject to abrasion.

3.5.3 Magnesium alloy parts - Wherever practicable, magnesium alloy parts shall be surface treated in accordance with MIL-M-3171 to provide protection against corrosion.

3.5.4 Plating - When used as a sole protective coating, cadmium plating shall be in accordance with QQ-P-416, Type I or II, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.6 Screw threads - Screw threads shall conform to the requirements of MIL-S-7742.

3.7 Safety wiring and staking - Accidental loosening of screws, screw parts, and other connections shall be prevented by safety wiring (0.032-inch minimum OD), staking, or other approved methods. Washers and cotter pins, where used, shall be assemblied in a manner which prevents rotation of washers and movement of cotter pins under conditions of vibration.

3.8 Performance - The blower shall satisfy the performance requirements specified in Section 4 when subjected to the following tests:

Heating	Transient voltage
Dielectric strength	Radio interference
Performance at sea level	Acceleration
Performance at altitude	Shock
Temperature rise	Vibration
Acoustical noise	Life
Ripple voltage	Humidity
Overspeed	Salt spray
Reverse polarity	Sand and dust
Operating position	Fungus
Starting current	Explosion-proof (when specified)

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3.9 Markings -

3.9.1 Terminals - Marking of terminals shall be in accordance with MIL-STD-130 unless otherwise shown by the applicable motor drawing, detail specification, or both. When deviations from MIL-STD-130 exist, a diagram or chart of the terminals shall be affixed to the motor.

3.9.2 Rotation marking - The direction of rotation of the motor rotor or fan blade shall be plainly and permanently indicated by an arrow.

3.9.3 Internal capacitors - The presence of internal capacitors which may be damaged by dielectric strength test of the motor shall be plainly and permanently indicated on the surface of the blower near the nameplate.

3.10 Identification of product - Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130. The identification data applied to the motor shall be as follows:

Nomenclature (as shown on applicable drawing)
For use on 28V dc system
Speed (RPM)
Amperes
Duty, Continuous
MS Part No.
Federal Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or Trade-Mark
Inspector's Stamp
US
Acceptance Date

Any additional nameplates with proprietary data shall be attached to the blower by means of easily removable screws.

3.10.1 Restriction on size of trademark - The manufacturer's name or trademark shall not be larger than other letters appearing on the nameplate.

3.10.2 Use of MIL designations - MIL designations shall not be applied to a product, except for qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification.

3.11 Workmanship - All machined surfaces shall have a finish suitable for the purpose intended, and all details of manufacture, including the preparation of

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parts and accessories, shall be in accordance with the best practice for high-quality electrical equipment. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts, plating, lacquering, riveting, clearance between soldered connections, removal of burrs and sharp edges, and ruggedness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests - The inspection and testing of a unit shall be classified as follows:

(a) Qualification tests - Qualification tests are those tests performed on samples submitted for approval as qualified products.

(b) Acceptance tests - Acceptance tests are those tests performed on individual lots which have been submitted for acceptance.

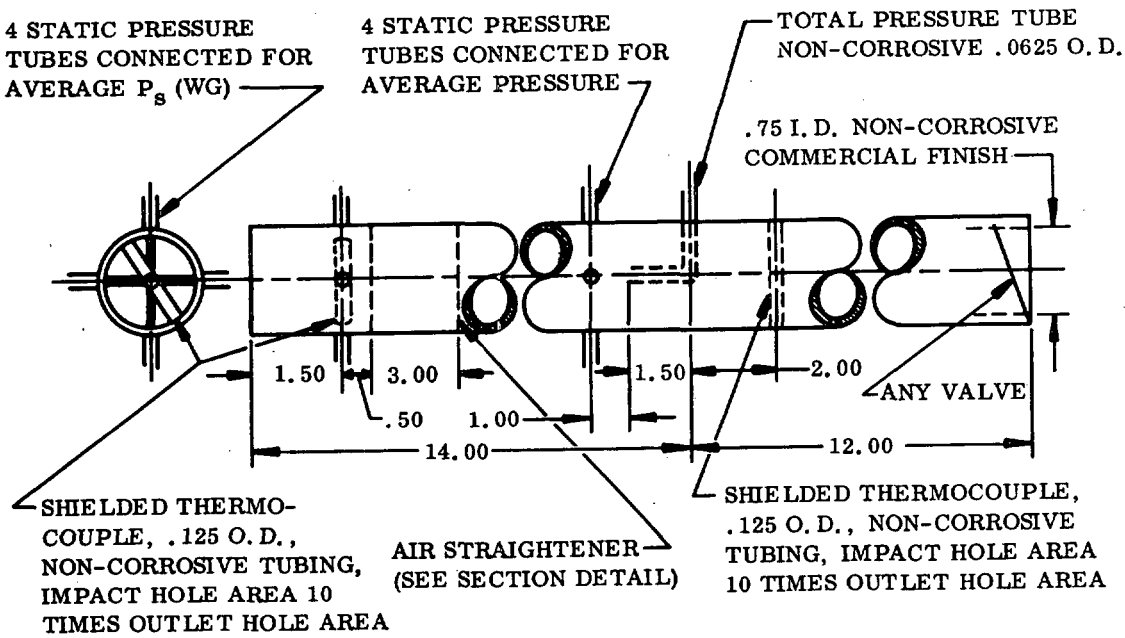
4.2 Qualification tests -

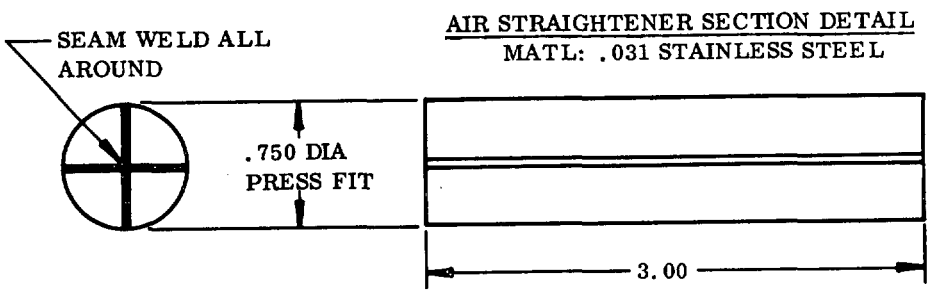
4.2.1 Sampling instructions - The qualification test samples shall consist of eight units upon which qualification is desired. The manufacturer shall furnish all samples and shall be responsible for accomplishing the required tests on four samples. Upon completion of the tests, the units tested, the four untested units, together with test reports, shall be identified as required and forwarded to the activity responsible for qualification as designated in the letter of authorization from that activity. The test reports shall include a statement that the samples submitted have been subjected to and have complied with Dielectric strength test (4.5.3). Units shall be accompanied by two copies of outline and detail assembly drawings thereof and by two copies of sample instructions with illustrations and diagram, if necessary, covering the installation of the units.

4.2.1.1 At the option of the procuring activity, any or all tests may be repeated. If the manufacturer desires to submit new units for test purposes, units which have passed tests at the manufacturer's plant shall accompany these new items.

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THE FOLLOWING METHOD SHALL BE USED TO MEASURE BLOWER OUTPUT





$$CFM = 1097\,A\sqrt{\frac{h_t - h_s}{D}}$$

$$D = \frac{1.325P}{T}$$

WHERE:

- A = PIPE AREA (SQ. FT)
- h_t = TOTAL PRESSURE (INCHES OF WATER)
- h_s = AVERAGE STATIC PRESSURE (INCHES OF WATER) STATIC PRESSURE SHALL BE MEASURED IN THE PLENUM FORWARD OF THE OUTLET. STATIC PRESSURE MAY BE REDUCED LINEARLY WITH ALTITUDE.
- P = ABSOLUTE AIR PRESSURE IN TUBE (INCHES OF MERCURY)
- T = $460 + T_1$ (°F)
- T_1 = AIR TEMPERATURE IN TUBE NEAR TOTAL PRESSURE TUBE

TOLERANCE FOR ALL DIMENSIONS IS ±.015

FIGURE 1

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4.2.2 Tests - The Qualification tests shall consist of all the tests of this specification, as described under "Test Methods." The tests shall be conducted on each of four units as shown below and must be conducted essentially in the order listed for each unit.

TESTS

UNIT NO. 1

	Para. No.
(a) Examination of product	4.5.1
(b) Heating	4.5.2
(c) Dielectric strength	4.5.3
(d) Performance at sea level	4.5.4
(e) Performance at altitude	4.5.5
(f) Temperature rise	4.5.6
(g) Acoustical noise	4.5.7
(h) Ripple voltage	4.5.8
(i) Overspeed	4.5.10
(j) Operating position	4.5.9
(k) Starting current	4.5.12
(l) Shock	4.5.16

UNIT NO. 2

	Para. No.
(a) Examination of product	4.5.1
(b) Heating	4.5.2
(c) Dielectric strength	4.5.3
(d) Performance at sea level	4.5.4
(e) Performance at altitude	4.5.5
(f) Temperature rise	4.5.6
(g) Acoustical noise	4.5.7
(h) Ripple voltage	4.5.8
(i) Overspeed	4.5.10
(j) Reverse polarity	4.5.11
(k) Operating position	4.5.9
(l) Starting current	4.5.12
(m) Transient voltage	4.5.13
(n) Radio interference	4.5.14
(o) Acceleration	4.5.15
(p) Humidity	4.5.19
(q) Explosion proof (When specified on applicable detail specification or drawing)	4.5.23

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TESTS (Cont'd)

UNIT NO. 3

		Para. No.
(a)	Examination of product	4.5.1
(b)	Heating	4.5.2
(c)	Dielectric strength	4.5.3
(d)	Performance at sea level	4.5.4
(e)	Temperature rise	4.5.6
(f)	Ripple voltage	4.5.8
(g)	Overspeed	4.5.10
(h)	Reverse polarity	4.5.11
(i)	Transient voltage	4.5.13
(j)	Radio interference	4.5.15
(k)	Vibration	4.5.17
(l)	Fungus	4.5.22
(m)	Salt spray	4.5.20

UNIT NO. 4

		Para. No.
(a)	Examination of product	4.5.1
(b)	Heating	4.5.2
(c)	Dielectric strength	4.5.3
(d)	Performance at sea level	4.5.4
(e)	Temperature rise	4.5.6
(f)	Ripple voltage	4.5.8
(g)	Overspeed	4.5.10
(h)	Life	4.5.18
(i)	Sand and dust	4.5.21

4.2.2.1 Disassembly and inspection - At the conclusion of the Qualification tests, the units will be disassembled and inspected for excessive wear and defects.

4.2.3 Rejection and retest of qualification samples - Units which have been rejected or returned to the manufacturer for any reason during Qualification tests, may be reworked or have parts replaced to correct defects. Before resubmitting the unit, full particulars concerning the rejection and the corrective action taken by the manufacturer must be submitted in writing by the manufacturer to the test activity and to the procuring activity. Tests shall not be resumed until such a report is received. Where Qualification tests are conducted under the auspices of the manufacturer, the Qualifying activity shall be advised upon failure of qualification sample and of the action taken by the manufacturer with regard to failure.

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4.3 Acceptance tests - Acceptance tests shall consist of Individual tests and Sampling tests.

4.3.1 Individual tests - Each blower shall be subjected to the following tests described under "Test Methods":

(a) Examination of Product (4.5.1)
(b) Dielectric Strength (4.5.3)
(c) Performance at Sea Level (4.5.4)
(d) Temperature Rise (4.5.6)
(e) Acoustical Noise (4.5.7)
(f) Ripple Voltage (4.5.8)
(g) Overspeed (4.5.10)
(h) Reverse Polarity (4.5.11)
(i) Transient Voltage (4.5.13)
(j) Radio Interference (4.5.14)

In addition, the blower shall be subject to any other tests specified herein which the Inspector, with the approval of the procuring agency, considers necessary to determine conformance with the requirements of this specification.

4.3.2 Sampling tests - Sampling tests shall be as specified by the procuring activity.

4.3.3 Rejection and retest - Blowers which have been rejected may be reworked or have parts replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the rejected unit shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.4 Test conditions - Unless otherwise specified, each test in this section shall be made under the following conditions.

4.4.1 Ambient - Tests specified shall be performed at an ambient temperature of 25°C ± 15°C.

4.4.2 Altitude - Tests will be run at approximately sea-level altitude.

4.4.3 Voltage measurement - The input voltage shall be measured at the terminals of the unit.

4.4.4 Input voltage - Nominal input voltage as shown on applicable drawings shall be used.

4.4.5 Position - The unit shall be mounted with its major axis horizontal and its base down.

4.4.6 Warmup - Prior to each test, the unit shall be operated at rated load for sufficient time to reach a substantially constant temperature, as defined in 4.5.2.

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4.5 Test methods -

4.5.1 Examination of product -

4.5.1.1 Qualification tests - Each unit subassembly and part shall be examined as the Inspector may deem necessary to determine compliance with this specification and the applicable drawing or detail specification with respect to material and workmanship, standard parts, simplification, lubrication, electrical connections, rotor removal, protection, threaded parts, finish, interchangeability, marking, instructions, dimensions, and weight.

4.5.1.2 Individual tests - Each unit subassembly and part shall be examined as the Inspector may deem necessary to determine compliance with this specification and the applicable drawing or detail specification with respect to material and workmanship, lubrication, electrical connections, threaded parts, finish, interchangeability, marking instructions, dimensions, and weight.

4.5.2 Heating - Provisions shall be made for determining speed, input voltage and current, field voltage and field current (where practicable), and the air flow. While the motor is cold, the resistance and temperature of the field shall be determined for use in calculating the field temperature rise (average) during operation at rated load. Where resistance method is not practicable, measurement may be made by a thermocouple on the frame or field structure. The blower shall be considered to have reached its continuous operating condition in the case of continuous-duty motors, when the rate of rise of field temperature above the then existing ambient is no more than 1°C for a period of 5 minutes.

4.5.2.1 Air Flow - The ability of the blower to deliver rated air flow at rated input voltage for continuous - or rated-duty cycle operation shall be demonstrated. Immediately following the above run, the ability of the motor to deliver the rated air flow for both minimum and maximum continuous input voltage shall be demonstrated. When specified by the detail specification or drawing, blower operation at reduced horsepower output for emergency or ground-start low-voltage limits of MIL-STD-704 shall be demonstrated. The temperature of exposed parts shall be limited to a maximum of 50°C and shall not be exceeded for any temperature-altitude condition within the requirements of this specification.

4.5.2.2 Speed - At rated voltage and rated air flow, the speed of the motor shall be determined. The speed of continuous-duty motors shall be within 10 percent of rated speed with the internal temperature of the motor as near as practicable to the temperature recorded during Heating test (4.5.2). When this test is conducted following environmental tests, the speed shall be within 15 percent of rated speed.

4.5.3 Dielectric strength - The motor windings shall be subjected to and withstand the following test voltage at a frequency of approximately 60 cps,

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applied between windings, and between each winding and frame, for the specified time. For this test, all windings permanently connected together are to be considered one winding.

500 V (rms) for 1 minute or
600 V (rms) for 1 second

This test shall be performed prior to the attachment of any electronic parts, such as diodes, capacitors, etc. No voltage breakdown shall occur as a result of this test.

4.5.3.1 Capacitors - Capacitors shall be subjected to and shall withstand for 1 second a test voltage of twice the maximum peak voltage encountered during normal operations.

4.5.4 Performance at sea level - The air output of the blower shall be measured with the equipment shown in Figure 1. The formula shown in Fig-1 shall be used to calculate airflow. During this test the unit shall be operated at or near sea level pressure. For each voltage condition the amount of output air shall not be less than the minimum value at the required static pressure shown on the applicable drawing.

4.5.4.1 Room temperature operation (25°C ±15°C) -

(a) At minimum input voltage the static pressure shall be adjusted to the minimum value required by the applicable drawing. The amount of air flow shall be calculated using the formula shown on Figure 1.

(b) At rated input voltage the preceding test shall be repeated.

(c) At maximum input voltage the preceding test shall be repeated.

(d) At the option of the qualifying service tests may be made at any other intermediate voltage to more accurately plot a curve of the output air.

4.5.4.2 Extreme temperature operation (-55°C) - Repeat 4.5.4.1 at -55°C.

4.5.4.3 Extreme temperature operation (+71°C) - Repeat 4.5.4.1 at +71°C.

4.5.5 Performance at altitude - For this test, rated input voltage shall be used. The blower shall be instrumented so that bearing, winding, stator and other hot spot or critical temperature readings can be checked and recorded. The blower shall be operated under the following altitude and temperature conditions required by Curve II of MS33543:

(a) One hour at standard test conditions followed by 6 hours at 10,000 feet altitude.

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(b) One hour at standard test conditions followed by 6 hours at 20,000 feet altitude.

(c) Two hours at 40,000 feet altitude. For this part the blower shall not be required to produce the amount of air required by the applicable drawing but to determine that the blower will not be damaged by operation at such altitude.

(d) The Altitude test shall consist of one cycle each under number (a), (b), and (c) above.

4.5.6 Temperature rise - During the course of tests under 4.5.4 and 4.5.5 the temperature increase of the air due to compression and energy dissipation through the blower shall not be more than the maximum allowed by 3.3.2.11.

4.5.7 Acoustical noise - The audible noise level shall be measured with a General Radio sound-level meter, Type 1551-B or equivalent meter. When measured within an area one foot from the unit, the acoustical noise shall not be more than the maximum specified by 3.3.15.

4.5.8 Ripple voltage - The ripple voltage shall be measured with a calibrated oscilloscope or with an Avtron ripple meter, Model T256 or equivalent. Peak voltage readings shall be taken with the voltmeter successively connected for each polarity. Unless otherwise specified by the applicable drawing or detail specification, the higher of the two readings shall not exceed 1.5 volts.

4.5.9 Operating position - During the course of other tests, it shall be ascertained that the operation of the blower in any position does not adversely affect power output, heating, lubrication, and other operating characteristics.

4.5.10 Overspeed - This test shall be made while the blower is hot as the result of testing and shall be made at no load and at an input voltage of 35 volts for five minutes.

4.5.11 Reverse polarity - Reverse rated voltage shall be applied to the input terminals for a period not less than 10 minutes. Following this test the unit shall meet the requirements of 4.5.4.1 (b).

4.5.12 Starting current - The starting line current shall be measured using a recording oscillograph while starting the blower at no load when at room ambient temperature; this line current shall not exceed 700 percent of the full load line current or 250 amperes, whichever is smaller. The line current shall return and stay within 300 percent of full load line current within 0.2 second after first application of input phase voltage. The current oscillogram shall indicate a definite current peak and shall have no tendency to remain at peak value.

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4.5.13 Transient voltage - The unit shall be subjected to the transient dc voltage as shown in MIL-STD-704, for category B utilization equipment, step 1. Following this test the unit shall meet the requirements of 4.5.4.1 (b).

4.5.14 Radio interference - At minimum, rated, and maximum input voltages, both conducted and radiated radio interference shall be measured using the test procedures and any applicable test instruments listed in MIL-I-6181. Units exceeding the interference limits set down by MIL-I-6181 shall be rejected.

4.5.15 Acceleration - The blower shall be mounted on a centrifuge. The centrifuge shall then be operated at a speed that will produce an acceleration of ten gravitational units; the blower shall then be rotated 180 degrees about an axis perpendicular to the axis of the centrifuge arm and the test repeated. The blower shall be mounted on the centrifuge in four additional positions such that the accelerating force is applied in both directions through each of the remaining axes of the blower. For each position of the blower, the centrifuge shall be operated at a speed that will produce an acceleration of 10 gravitational units. The acceleration of 10 gravitational units in each test shall be stabilized and maintained for a period of not less than 1 minute..

4.5.16 Shock - The unit shall be subjected to the Shock tests of MIL-E-5272, Procedure V, Equipment Operation. Damage or failure of any part of the unit during this test shall be cause for rejection. Following these tests, the unit shall pass the test specified in 4.5.4.1 (b) of this specification. The unit shall then be subjected to the Shock tests of MIL-E-5272, Procedure V, Equipment Crash Safety.

4.5.17 Vibration - The unit shall be subjected to the Vibration tests, Procedure XII of MIL-E-5272. The blower shall be operated at free delivery during the entire test. Any mechanical or electrical damage occurring as a result of the Vibration test shall be cause for rejection.

4.5.18 Life - The blower shall operate for 1500 hours at rated voltage in an ambient temperature of 25°C ± 15°C. The output air shall be restricted sufficiently to produce, forward of the restriction, the required static pressure shown on the applicable drawing. The unit shall be brought to a complete stop and started within 3 minutes at least once every 25 hours. No maintenance of any kind shall be allowed during this test.

4.5.19 Humidity - The unit shall be subjected to the Humidity tests, Procedure I of MIL-E-5272. Immediately following this test, the unit shall pass the test specified in 4.5.4.1 (b).

4.5.20 Salt spray - The unit shall be subjected to 25 hours of salt spray in accordance with the Salt spray tests of MIL-E-5272. Following this test, the unit shall be washed, and dried for 15 to 20 hours, and then shall meet the requirements of 4.5.4.1 (b). There shall be no evidence of excessive corrosion.

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4.5.21 Sand and dust - The unit shall be subjected to the Sand and dust tests, Procedure I of MIL-E-5272 at sand and dust velocities of 200 ± 100 fpm. The 8-hour test cycle shall be conducted with blower inoperative. At the end of the test, the unit shall be removed from the test chamber and subjected to the test specified in 4.5.4.1 (b).

4.5.22 Fungus - The unit shall be subjected to the Fungus resistance tests, Procedure I, of MIL-E-5272. Immediately following this test, the unit shall meet the test requirements of 4.5.4.1 (b).

4.5.23 Explosion-proof - Explosion-proof motors shall be subjected to Explosion-proof tests, Procedure IV of MIL-E-5272. These tests shall be made using two different explosive mixtures; one mixture resulting in maximum pressure and the other in maximum duration of flame. Tests shall be conducted at sea level only; five explosions shall be accomplished for each mixture. The explosive mixture may be circulated within the chamber and motor at sea-level pressure in lieu of 10,000 feet as specified for single tests, under Procedure III of subject test in MIL-E-5272. Motors below 2 inches in diameter may be tested in accordance with Procedure III in lieu of Procedure IV.

4.6 Material test specimens - Test specimens shall be furnished when requested by the Government for check tests at a Government laboratory to determine conformance with the applicable specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - The blowers shall be unit packaged in accordance with MIL-P-7936, Level A or C as specified in the contract or order.

5.1.1 Level A - For Level A packaging, preservation shall be in accordance with MIL-P-116, Method II, without preservative compounds, using metal reusable containers, unless otherwise specified.

5.1.2 Level C - For Level C the blowers shall be preserved and packaged individually in accordance with manufacturer's commercial practice.

5.2 Packing - Packing shall be in accordance with MIL-P-7936, Level A, B, or C as specified in the contract or order.

5.3 Marking - Interior and exterior containers shall be marked in accordance with MIL-P-7936 and MIL-STD-129.

5.3.1 Reinspection dates - Reinspection date marking shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use - This specification covers the requirements of dc blowers for use in aircraft 28 volt dc systems to ventilate exposure suits.

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6.2 Ordering data - Procurement documents should specify the following:

(a) MS part number or title, number, and date of detail specification.

(b) Selection of applicable levels of preservation and packaging, and packing.

6.3 Provisions for qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command AIR 53032, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity.

6.4 Definitions -

6.4.1 Ambient temperature - Ambient temperature is the temperature of air surrounding the unit in normal free convection (velocity below 50 fpm).

6.4.2 Continuous duty - Continuous duty is a requirement of service that demands operation at substantially constant load for an indefinite period.

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